

General Waveform Decomposition for Neural Network Emulation

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Abstract

We present a general framework for neural network emulation of gravitational waveforms that can accommodate arbitrary physical complexity: precession, higher modes, and eccentricity. We clarify which aspects of the standard decomposition are coordinate choices versus restrictive physical assumptions, and propose a neural network architecture that learns the optimal frame decomposition end-to-end without requiring external Euler angle extraction.

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1 Introduction

Existing neural network emulators for gravitational waveforms make simplifying assumptions that limit their generality:

- **mlgw_NN** [1]: Assumes aligned spins (non-precessing) and circular orbits
- **SEOBNN** [2]: Assumes single-spin systems ($\vec{\chi}_2 = 0$), conjugate mode symmetry in the coprecessing frame, and circular orbits
- **SEOBNRE_AI** [3]: Includes eccentricity but only models the dominant $(2, 2)$ mode with aligned spins

This document presents a framework that can accommodate any level of complexity, and addresses practical concerns about deploying such a framework on arbitrary waveform models.

2 General Decomposition

2.1 Spherical Harmonic Expansion

The gravitational wave strain observed at angles (ι, φ_0) from the source is expanded in spin-weighted spherical harmonics:

$$h(t; \boldsymbol{\lambda}; \iota, \varphi_0) = \sum_{\ell=2}^{\ell_{\max}} \sum_{m=-\ell}^{+\ell} h_{\ell m}(t; \boldsymbol{\lambda}) {}^{-2}Y_{\ell m}(\iota, \varphi_0) \quad (1)$$

where ${}^{-2}Y_{\ell m}$ are the spin -2 weighted spherical harmonics, and $\boldsymbol{\lambda}$ denotes the *intrinsic* parameter space.

2.2 Intrinsic vs Extrinsic Parameters

A crucial feature of the decomposition (1) is that it separates parameters into two classes:

Intrinsic parameters $\boldsymbol{\lambda}$: Properties of the source that affect the mode functions $h_{\ell m}(t)$:

$$\boldsymbol{\lambda} = \{q, \chi_{1x}, \chi_{1y}, \chi_{1z}, \chi_{2x}, \chi_{2y}, \chi_{2z}, e_0, \dots\} \quad (2)$$

where $q = m_1/m_2$ is the mass ratio, $\vec{\chi}_1, \vec{\chi}_2$ are the dimensionless spin vectors, and e_0 is the orbital eccentricity.

Extrinsic parameters: Properties that affect only how the modes combine or scale:

- **Inclination ι :** Enters only through ${}^{-2}Y_{\ell m}(\iota, \varphi_0)$
- **Reference phase φ_0 :** Enters only through ${}^{-2}Y_{\ell m}(\iota, \varphi_0)$
- **Luminosity distance d_L :** Scales amplitude as $h \propto 1/d_L$
- **Polarisation angle ψ :** Rotates $h_+, h_\times$ at the detector

Implication for emulation: The neural network only needs to learn the intrinsic modes $h_{\ell m}(t; \boldsymbol{\lambda})$. Extrinsic parameters are handled analytically at inference time via the known spherical harmonic functions. One trained model works for all viewing angles.

2.3 Coprecessing Frame Decomposition

For precessing systems, we decompose each inertial-frame mode into coprecessing-frame modes via a time-dependent rotation:

$$h_{\ell m}(t; \boldsymbol{\lambda}) = \sum_{m'=-\ell}^{+\ell} D_{mm'}^{\ell}(\alpha(t), \beta(t), \gamma(t); \boldsymbol{\lambda}) h_{\ell m'}^{\text{cp}}(t; \boldsymbol{\lambda}) \quad (3)$$

where $D_{mm'}^{\ell}$ are the Wigner D-matrices parameterised by Euler angles $\{\alpha(t), \beta(t), \gamma(t)\}$ that track the evolution of the orbital angular momentum $\hat{L}(t)$.

2.4 Amplitude–Phase Decomposition

Each coprecessing mode is decomposed into amplitude and phase:

$$h_{\ell m}^{\text{cp}}(t; \boldsymbol{\lambda}) = A_{\ell m}(t; \boldsymbol{\lambda}) e^{i\phi_{\ell m}(t; \boldsymbol{\lambda})} \quad (4)$$

This decomposition is advantageous because $A_{\ell m}(t)$ and $\phi_{\ell m}(t)$ are smooth functions that are easier for neural networks to learn than the oscillatory complex waveform.

3 What is a Coordinate Choice vs a Restrictive Assumption

3.1 Coordinate Choices (Not Restrictive)

The following are *coordinate transformations* that can be applied to any waveform without loss of generality:

1. **Spherical harmonic expansion** (1): Always valid for any radiation field.
2. **Coprecessing frame rotation** (3): This is just a time-dependent rotation of coordinates. The coprecessing frame is defined by tracking $\hat{L}(t)$, which exists for any binary orbit. The transformation is invertible and lossless.
3. **Amplitude–phase decomposition** (4): Any complex function can be written in polar form.

3.2 Restrictive Assumptions

The following are *physical approximations* that limit the class of waveforms that can be accurately represented:

1. **Conjugate symmetry in the inertial frame** (aligned-spin assumption):

$$h_{\ell, -m}(t; \boldsymbol{\lambda}) = (-1)^{\ell} h_{\ell m}^{*}(t; \boldsymbol{\lambda}) \quad (5)$$

This holds only for non-precessing systems with aligned spins.

2. **Conjugate symmetry in the coprecessing frame:**

$$h_{\ell, -m}^{\text{cp}}(t; \boldsymbol{\lambda}) = (-1)^{\ell} (h_{\ell m}^{\text{cp}}(t; \boldsymbol{\lambda}))^{*} \quad (6)$$

This approximation assumes the coprecessing-frame waveform looks like an aligned-spin waveform. It works well for mild precession but breaks down for strongly precessing systems.

3. **Single-spin assumption** ($\vec{\chi}_2 = 0$): Reduces the parameter space but excludes double-spin systems.
4. **Circular orbit assumption** ($e_0 = 0$): Excludes eccentric binaries.

3.3 Summary Table

Component	Type	Always Valid?
Spherical harmonic expansion	Coordinate choice	Yes
Coprecessing frame rotation	Coordinate choice	Yes
Amplitude/phase split	Coordinate choice	Yes
Conjugate symmetry (inertial)	Approximation	No (aligned spin only)
Conjugate symmetry (coprecessing)	Approximation	No (mild precession only)
Single spin	Approximation	No
Circular orbits	Approximation	No

Table 1: Classification of decomposition components.

4 Where Different Physics Enters

4.1 Precession

Precession is handled by the coprecessing frame transformation. The Euler angles $\alpha(t), \beta(t), \gamma(t)$ encode the precession dynamics—the time-dependent orientation of the orbital plane.

Key point: The coprecessing frame transformation *handles* precession but doesn't *add* it. If the underlying modes don't contain precession physics, the transformation is trivial ($D_{mm'}^\ell \rightarrow \delta_{mm'}$).

4.2 Eccentricity

Eccentricity affects the *intrinsic* time evolution of the modes. It enters through:

1. Mode amplitudes $A_{\ell m}(t)$: Eccentric orbits produce amplitude modulations at the orbital frequency
2. Mode phases $\phi_{\ell m}(t)$: More complex orbital phase evolution with periastron advance
3. Euler angles $\alpha(t), \beta(t), \gamma(t)$: Precession dynamics couple to eccentricity
4. The coprecessing frame definition: $\hat{L}(t)$ has more complex trajectory

Key point: Eccentricity cannot be added by coordinate transformations. It must be present in the underlying waveform model being emulated.

4.3 Implications for IMRPhenomX Family

- **IMRPhenomXHM**: Non-precessing, higher modes, circular. Can serve as coprecessing modes.
- **IMRPhenomXPHM**: Uses XHM as coprecessing modes + Euler angle rotation. Precessing, higher modes, circular.
- **Neither has eccentricity**: To emulate eccentric waveforms, one must start from an eccentric model (e.g., SEOBNRE, TEOBResumS-GIOTTO).

5 The Euler Angle Extraction Problem

5.1 The Practical Challenge

For emulation, we assume the input is a “black box” waveform code that outputs only inertial-frame quantities:

$$\text{Code : } \lambda \mapsto \{h_+(t), h_\times(t)\} \quad \text{or} \quad \lambda \mapsto \{h_{\ell m}(t)\} \quad (7)$$

The copreprocessing modes and Euler angles are *not* provided. Some codes (e.g., SEOB-NRv4PHM in LALSuite) do expose these internally, but this cannot be assumed in general.

5.2 Non-Uniqueness of the Copreprocessing Frame

There is no single “correct” definition of the copreprocessing frame. Valid choices include:

- ***L*-frame:** Track instantaneous orbital angular momentum $\hat{L}(t)$
- ***J*-frame:** Align with total angular momentum $\hat{J}$ at reference time
- **Quadrupole-preferred frame:** Minimise power in non- $(2, \pm 2)$ modes
- **Minimal rotation frame:** Impose $\dot{\gamma} = -\dot{\alpha} \cos \beta$

Different conventions give different Euler angles and copreprocessing modes, though all are valid coordinate choices.

5.3 Classical Extraction Methods

If one chooses to extract Euler angles from inertial-frame waveforms, established methods exist:

1. **Dominant eigenvector method:** Find the frame maximising power in $(2, \pm 2)$
2. **Phase gradient method:** Extract from the $(2, 2)$ mode phase evolution
3. **Optimisation:** Minimise some cost function over rotation parameters

These methods work but introduce preprocessing complexity and convention dependence.

6 Neural Network Architecture Options

We consider three architectural approaches for learning precessing waveforms.

6.1 Option 1: Extract Then Learn

1. Use classical algorithms to extract Euler angles from training waveforms
2. Train separate neural networks for copreprocessing modes and Euler angles
3. At inference, predict both and combine via Wigner rotation

Pros: Physics-informed, interpretable, follows literature

Cons: Extraction is non-unique, adds preprocessing step, convention-dependent

6.2 Option 2: Learn Inertial Modes Directly

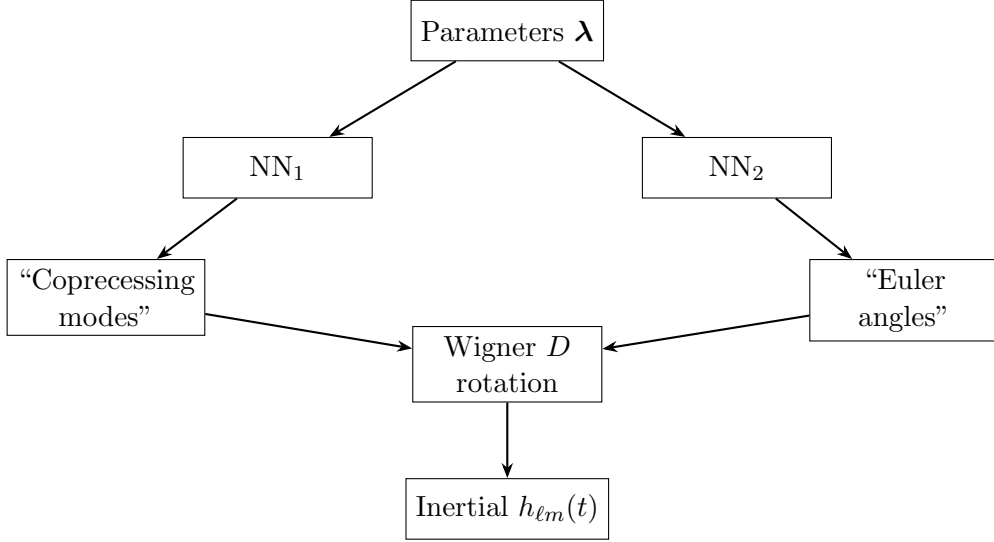
1. Skip decomposition entirely
2. Train neural network: $\lambda \mapsto h_{\ell m}(t)$ in inertial frame

Pros: Simple pipeline, no extraction needed

Cons: Inertial modes have fast precession oscillations, much harder to learn, requires larger networks

6.3 Option 3: Learn Frame Decomposition End-to-End (Recommended)

Build the Wigner rotation into the network architecture:



The loss function is computed on the inertial-frame output:

$$\mathcal{L} = \sum_{\ell m} \left\| h_{\ell m}^{\text{pred}}(t) - h_{\ell m}^{\text{true}}(t) \right\|^2 \quad (8)$$

Key features:

- No external Euler angle extraction needed
- Physics structure (rotation equivariance) is built into architecture
- Network learns whatever internal decomposition works best
- The learned “Euler angles” are latent variables—they need not match any standard convention
- Works with any code that outputs $h_{\ell m}(t)$ in the inertial frame

Pros: General, no preprocessing, physics-informed architecture

Cons: More complex architecture, learned angles may not be interpretable

6.4 Comparison

7 Architecture Requirements

7.1 Without Conjugate Symmetry

For a fully general emulator covering $\ell \leq \ell_{\text{max}}$ without assuming conjugate symmetry:

For $\ell_{\text{max}} = 4$ (modes up to $(4, \pm 4)$): $5 + 7 + 9 = 21$ complex modes $\times 2$ (amp/phase) + 3 Euler angles = 45 output functions.

Criterion	Option 1	Option 2	Option 3
External extraction needed	Yes	No	No
Convention dependence	Yes	No	No
Physics structure exploited	Yes	No	Yes
Interpretable intermediates	Yes	N/A	No
Network complexity	Medium	High	Medium
Training difficulty	Medium	Hard	Medium

Table 2: Comparison of architectural options.

Component	Count	Notes
Coprocessing amplitudes $A_{\ell m}(t)$	$\sum_{\ell=2}^{\ell_{\max}} (2\ell + 1)$	All $m \in [-\ell, +\ell]$
Coprocessing phases $\phi_{\ell m}(t)$	$\sum_{\ell=2}^{\ell_{\max}} (2\ell + 1)$	All $m \in [-\ell, +\ell]$
Euler angles	3	$\alpha(t), \beta(t), \gamma(t)$

Table 3: Neural network outputs required for general emulation.

7.2 With Conjugate Symmetry (if applicable)

If conjugate symmetry can be assumed (aligned spins or mild precession), only $m \geq 0$ modes need to be modelled, roughly halving the outputs.

8 Summary

The complete general decomposition is:

$$h(t; \boldsymbol{\lambda}; \iota, \varphi_0) = \sum_{\ell, m} h_{\ell m}(t; \boldsymbol{\lambda})^{-2} Y_{\ell m}(\iota, \varphi_0) \quad (9)$$

$$h_{\ell m}(t; \boldsymbol{\lambda}) = \sum_{m'} D_{mm'}^{\ell}(\alpha, \beta, \gamma; t, \boldsymbol{\lambda}) h_{\ell m'}^{\text{cp}}(t; \boldsymbol{\lambda}) \quad (10)$$

$$h_{\ell m}^{\text{cp}}(t; \boldsymbol{\lambda}) = A_{\ell m}(t; \boldsymbol{\lambda}) e^{i\phi_{\ell m}(t; \boldsymbol{\lambda})} \quad (11)$$

Key points:

1. Inclination and reference phase are extrinsic—handled analytically via spherical harmonics.
2. The coprocessing frame decomposition is a coordinate choice, not a restrictive assumption.
3. Conjugate symmetry (in either frame) IS a restrictive assumption.
4. Eccentricity must be in the underlying model—it cannot be added by coordinate transformations.
5. For practical emulation, we recommend learning the frame decomposition end-to-end (Option 3), avoiding the need for external Euler angle extraction.

References

- [1] T. Grimbergen, S. Schmidt, C. Kalaghatgi, C. Van Den Broeck, “Generating Higher Order Modes from Binary Black Hole mergers with Machine Learning,” arXiv:2402.06587 (2024).

- [2] L. M. Thomas, G. Pratten, P. Schmidt, “Accelerating multimodal gravitational waveforms from precessing compact binaries with artificial neural networks,” arXiv:2205.14066 (2022).
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